

Scaling Laws

DATA 443/643 · Week 7, note 7.1

Week 7

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Ideas from Stanford CS224N 2025 L9 (Kaplan curves; GPT-3 tokens vs Chinchilla). Original slides; that course is not copied.

Lecture plan

- 1 Loss as a power of compute
- 2 A numeric on $L(C)$
- 3 Chinchilla: tokens with parameters
- 4 What a law does not tell you

Key goal: write N and D , not only $7B$. L is pretraining CE, not chat quality.

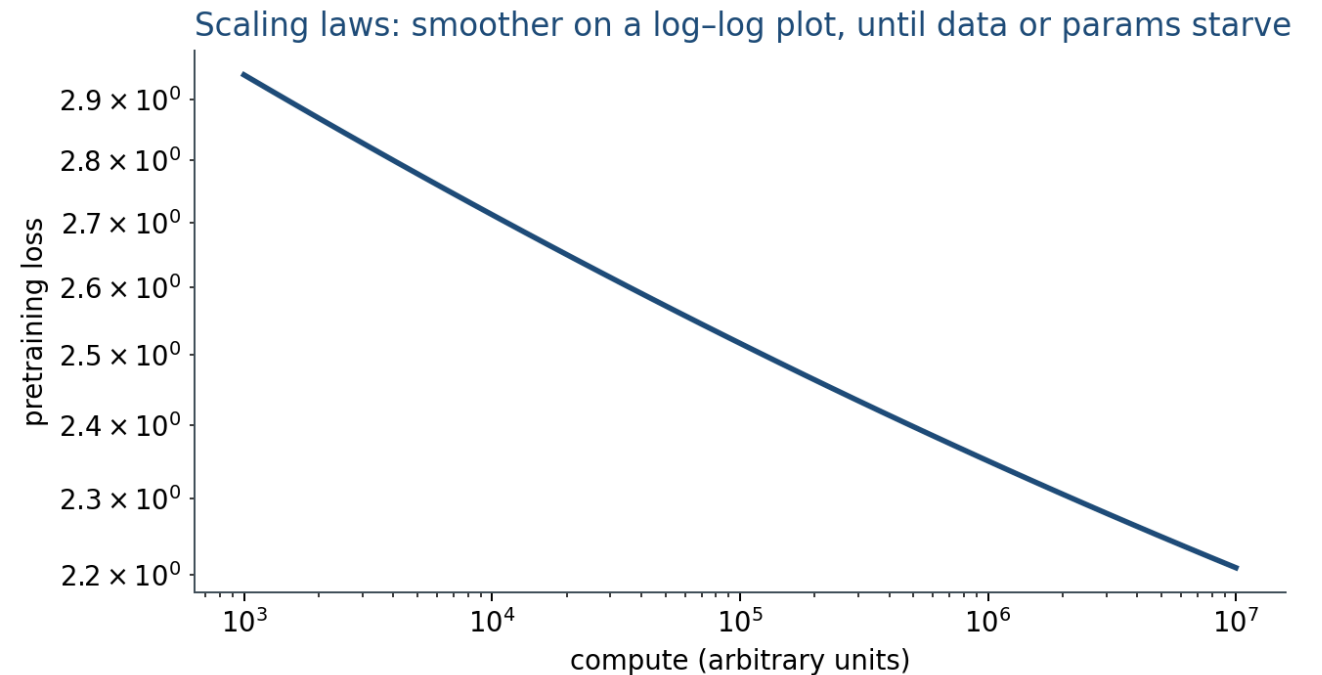
First time: what / why / formula

- What / why. Kaplan: loss vs compute (and N, D) looks like a power law plus a floor. Chinchilla: at fixed FLOPs, grow tokens and parameters together (~ 20 tokens/parameter in that paper). “We trained a 7B” is not a complete sentence. You need the token budget and whether you were compute-optimal.
- Architecture. Not a new net. A log–log plot and a recipe (N, D, C) .
- Formula. $L(C) \approx aC^{-b} + L_\infty$. Training FLOPs cartoon: $\approx 6ND$.
- Tradeoffs. + Planning a run. – L is not chat quality; data mixture and alignment sit off the curve; four toy points are not a law.

How. Fit $L \approx aC^{-b} + L_\infty$ (Lab 7 algebra). Then do Chinchilla arithmetic: $7B \times 20 = 140B$ tokens.

A smoother log-log curve

- $L(C) \approx aC^{-b} + L_{\infty}$.
- Extra FLOPs still buy loss, with diminishing returns.
- Lab 7 fits a , b , c on toy points. The paper's range is many decades.



One hundred times the compute

$$L = 2 C^{-0.15} + 1.4$$

$$C = 10^4: L \approx 1.90.$$

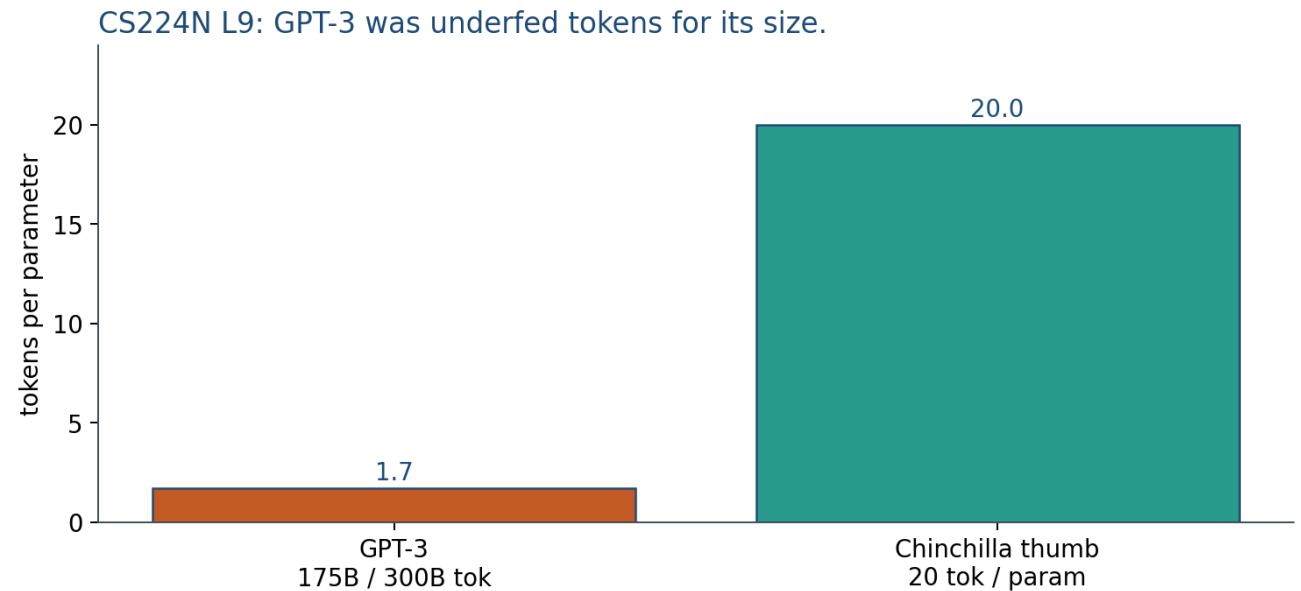
$$C = 10^6: L \approx 1.65. \text{ A}$$

100× jump bought 0.25 nats.

If you already sit on L_∞ , extra FLOPs buy almost nothing.

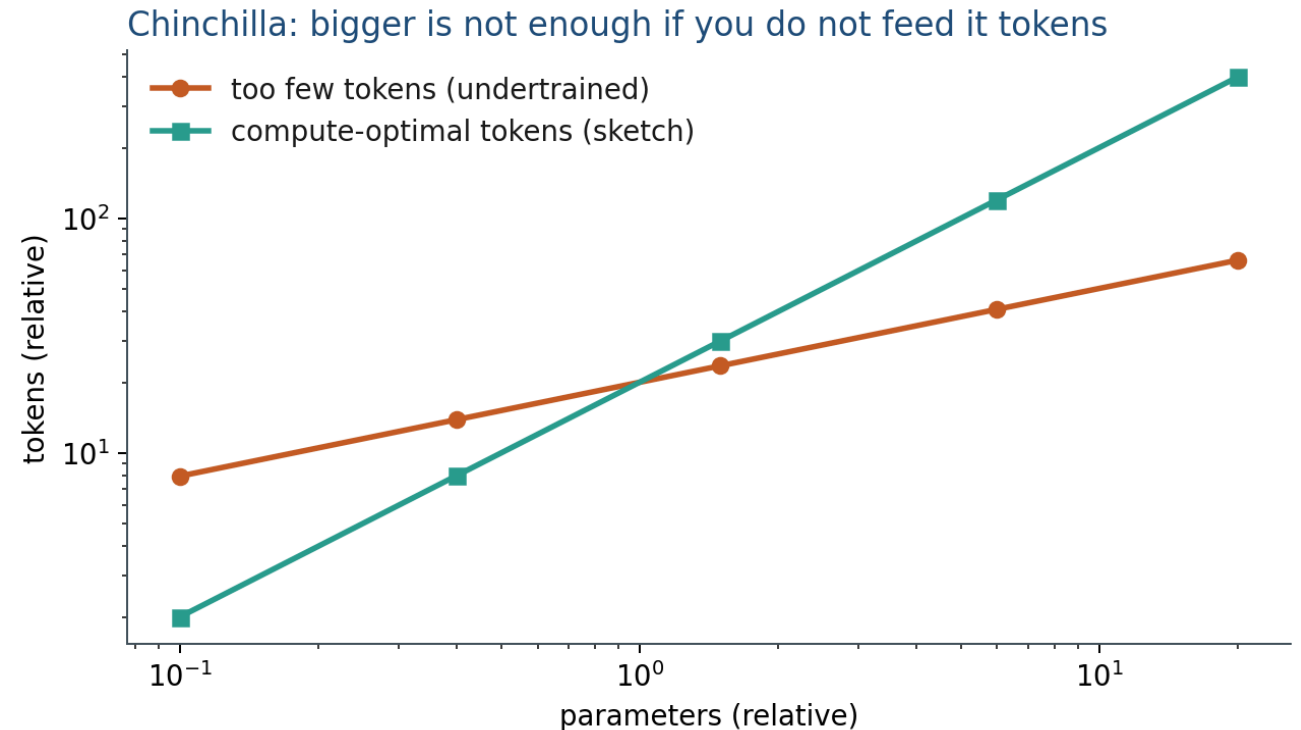
GPT-3 was underfed

- CS224N L9: 175B parameters, 300B tokens.
- That is about 1.7 tokens per parameter, not 20.
- A 70B trained with enough tokens beat larger starved nets.



Same FLOPs, different (N, D)

- Training FLOPs scale like $6ND$.
- A: $4N_0$ params, $D_0 / 4$ tokens. B: N_0 , D_0 .
- Chinchilla picks the balanced pair, not the wide starved net.



What to take from this lecture

Write D

7B × 20 tokens/param wants 140B tokens.

L is not chat

Alignment, mixture, and eval sit off the curve.

Lab 7

Fit a, b, c. Four noisy points are not a law.